

Transparent Statistical Learning with R/Shiny: An Interactive Platform for Discriminant Analysis

Eduardo Gonzalo Moyano Baquero

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Executive Summary

Transparent Statistical Learning with R/Shiny is an educational and interactive platform developed in R/Shiny, designed to integrate statistical theory, data preparation, diagnostics, methodological selection, modeling, evaluation, and result interpretation into a unified workflow. Discriminant Analysis is the platform's methodological foundation and primary focus. Its application requires coordinating various procedures, packages, and decision criteria available in R. The platform structures the learning and analysis process according to the sequence theory $\rightarrow$ data $\rightarrow$ diagnostics $\rightarrow$ methodological decision $\rightarrow$ model $\rightarrow$ results $\rightarrow$ evaluation $\rightarrow$ interpretation.

The platform currently has a functional MVP organized into eight tabs: Load Data, Exploratory Analysis, Missing Values, Categorical Variables, Models, Discriminant Analysis, Logistic Regression, and Help. The platform incorporates LDA/QDA, logistic regression, classical and modern decision trees, and Random Forest, with tables, metrics, visualizations, and explanations to facilitate interpretation of the results.

The first funded phase will last six months and will focus on the technical strengthening of the MVP, with emphasis on stability, modularization, reproducibility, testing, statistical verification, visualizations, and interface. The objective will be to strengthen existing functionalities and improve the robustness and reusability of the platform through statistical tools and procedures available in R. The development of new statistical algorithms is not included.

The funded work will be carried out in two three-month stages: stabilization and preparation, followed by strengthening and improvements. The phase will include automated testing, QA, and verification of Discriminant Analysis results using reference datasets, controlled cases, and independent calculations. As a result, the project will deliver a technically strengthened, stable, tested, and reproducible MVP, with modular code, improved priority functionalities, and strengthened visualizations. Subsequent phases may address comprehensive statistical and functional validation, expanded documentation, publication, translation, deployment, and other consolidation activities

The project has approximately 720 hours of prior work, valued at US\$21,600. The funded phase includes 333 hours, equivalent to US\$9,990. The prior work constitutes a contribution to the project and is not part of the requested funding.

Signatories

Project team

Eduardo Gonzalo Moyano Baquero, M.A.E. Principal Investigator and Lead Developer

Engineer specializing in Statistics and Computer Science. Responsible for the methodological design, architecture and implementation in R/Shiny, development of the Discriminant Analysis workflows, integration of statistical procedures, and implementation of automated testing and statistical verification processes. Approximately six months of prior prototyping and development work provide the technical foundation for the present funded phase.

Live Interactive Prototype (MVP):

<https://bds-eduardo-baquero.shinyapps.io/SOFTWARE-DISCRIMINANTE/>

Lilia Beatriz Santos Díaz, M.A.E., M.S.I.G., M.I.A. UX/UI and Front-End Specialist

Computer systems engineer with experience in enterprise software development. Responsible for the design and evaluation of the user experience and user interface (UX/UI), development and review of visualizations, usability evaluation, and execution of interaction and application functionality tests.

Ma. Mercedes Jaramillo Campos, M.A.E. QA, Process, and Documentation Specialist

Computer systems engineer with experience in process analysis and documentation. Responsible for quality assurance (QA), planning, execution, and documentation of tests, incident tracking and management, regression testing, technical documentation, preparation of the user manual, and consolidation of project deliverables.

Consulted

No formal external consultations have been conducted regarding this proposal.

The Problem

Applying Discriminant Analysis requires integrating several stages, including data preparation and exploration, assessment of assumptions, application of statistical tests, selection of procedures, model fitting, validation, and interpretation of results. Although R provides numerous functions and packages for performing these tasks, users must identify and coordinate the relevant tools, adapt the code to the characteristics of each dataset, and integrate the results obtained across the different stages of the analysis. The project addresses the difficulty of applying Discriminant Analysis in an integrated, reproducible, and transparent manner while coordinating the statistical tools available in R and facilitating the connection between diagnostics, methodological decisions, modeling, validation, and interpretation of results. Users who apply Discriminant Analysis, regardless of their level of experience with R.

The Transparent Statistical Learning with R/Shiny platform seeks to reduce the time and effort required to perform these analyses by integrating existing R tools into a simpler, understandable, reproducible, and transparent workflow, enabling users to better leverage the statistical capabilities of the language. The workflow explicitly maintains the relationship between the main stages of the analysis: data $\rightarrow$ diagnostics $\rightarrow$ methodological decision $\rightarrow$ modeling $\rightarrow$ validation $\rightarrow$ interpretation, facilitating an understanding of the decisions made and the traceability of the results obtained. The project has approximately six months of prior work in prototyping and development, during which a functional MVP was built in R/Shiny. The MVP integrates the main stages of the statistical analysis and constitutes the technical foundation of the present proposal.

Funding will be used to consolidate and strengthen the MVP through improvements in stability, modularization, reproducibility, testing, statistical validation, interface and visualizations. The development of new statistical algorithms is outside the scope of this project; the objective is to strengthen the integration and transparent presentation of existing statistical procedures in R, increasing the robustness and reusability of the platform. The project does not require modifications to the R core. Its objective is to integrate and facilitate

the use of existing functions, packages, and statistical procedures in R through a structured, understandable, and traceable workflow.

The proposal

Overview

Transparent Statistical Learning with R/Shiny proposes consolidating a functional MVP designed to facilitate the application of Discriminant Analysis through an integrated workflow that connects data preparation, statistical diagnostics, methodological selection, modeling, evaluation, and interpretation of results.

Funding will support the technical strengthening of the MVP over six months, with emphasis on stability, modularization, reproducibility, testing, statistical verification, visualizations, and interface, strengthening existing functionalities and improving the robustness, reproducibility, and reusability of the platform through statistical tools and procedures available in R. The funded work will be carried out in two three-month stages: stabilization and preparation, followed by strengthening and improvements. These activities will provide a stronger and more reusable R/Shiny platform for users applying Discriminant Analysis.

Detail

Minimum Viable Product

The current MVP is a functional application developed in R/Shiny that integrates the main stages of statistical analysis within an interactive environment. The platform was initially developed around Discriminant Analysis, which constitutes the methodological focus of the platform, and subsequently incorporated other supervised statistical modeling alternatives.

The application is currently organized into eight main tabs: Load Data, Exploratory Analysis, Missing Values, Categorical Variables, Models, Discriminant Analysis, Logistic Regression, and Help. The MVP interface is currently available in Spanish. The platform provides more comprehensive analyses for the following models due to the assumptions, diagnostic criteria, and statistical procedures involved in their application:

- **Discriminant Analysis:** The platform evaluates multivariate normality, means, standard deviations, statistical tests, covariance matrices, and Wilks' lambda. Based on these results, it provides diagnostic information and guidance for selecting between Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA).
- **Logistic Regression:** The platform supports

binomial and multinomial models and includes model-fit and performance indicators, multicollinearity analysis using VIF, and the detection of potentially influential observations using Cook's distance.

For all available models, the platform generates results and evaluation elements that include confusion matrices, performance metrics, and ROC curves. Depending on the model, additional outputs are provided, such as coefficient matrices, rule tables, and visualizations including decision trees, variable importance, and analysis of potentially influential observations. The platform also allows users to copy generated tables and charts for use in reports and documentation.

Figure 1 summarizes the main functionalities currently available in the MVP. The Discriminant Analysis and Logistic Regression options, due to their methodological relevance and the specific statistical requirements involved, are described in greater detail above.

Tab	Current Functionality
1. Load Data	Loads Excel files, validates the format, and displays data and instructions.
2. Exploratory Analysis	Displays percentage tables; allows users to select variables for imputation or proceed directly to Categorical Variables when no variables are selected.
3. Missing Values	Allows missing values to be handled according to variable type; includes visualizations and Save Imputation and Continue buttons.
4. Categorical Variables	Allows users to select variables, transform their values, convert them to factors, or perform both actions, with execution
5. Models	Allows users to select the dependent and independent variables, the training/test split, and five modeling alternatives.
8. Help	Provides information on models, assumptions, diagnostics, metrics, and interpretation.

Figure 1: Summary of Current Functionalities

The specific MVP improvements are presented in the functional and non-functional requirements sections.

Functional Requirements

- FR01. Improve the interface and interaction flows: Optimize navigation, organization, and analysis configuration through the use of Shiny and bslib, promoting a clearer and more consistent user experience.
- FR02. Improve visualizations: Enhance the presentation and interactivity of charts and results to facilitate their exploration and interpretation.
- FR03. Strengthen existing models: improve modeling procedures, diagnostics, results, and visualizations while maintaining the statistical procedures available in R as their foundation.

Non-Functional Requirements

NFR01. Modularity and maintainability: Consolidate modular and reusable code, separating the user interface, processing, statistical analysis, and visualization to facilitate maintenance.

NFR02. Quality and testing: Implement automated testing in R to verify core functions and critical components and to detect regressions during further development. NFR03. Statistical verification: Verify the results of tab 6 (Discriminant Analysis) using reference datasets, controlled cases, and independent calculations. NFR04. Usability: Improve the organization, navigation, consistency, and clarity of the interface, facilitating the execution of analyses and interpretation of the results obtained. NFR05. Reproducibility: Maintain an organized, version-controlled repository and manage dependencies through renv to facilitate consistent reproduction of the environment and analyses.

Architecture

Transparent Statistical Learning with R/Shiny is developed as an interactive application in R/Shiny, where R serves as the processing and statistical analysis engine, while Shiny provides the web environment, user interface, and management of the application's interaction flows and reactivity. The current codebase includes separate functions for data processing, statistical procedures, and the generation of tables and visualizations. The project will consolidate this structure through greater separation between reusable functions and user interface logic, together with improvements to reactive workflows and reproducibility.

Assumptions

- The architecture based on R/Shiny will be maintained, using the current MVP as the foundation for the funded phase.
- Statistical functionalities will primarily use existing packages and procedures in R, without developing new statistical algorithms.
- Interface and visualization improvements will be prioritized according to their contribution to clarity, navigation, understanding, and interpretation of results.
- Functionalities requiring research or technical evaluation will be analyzed and tested before their final incorporation.

External dependencies

The project depends on the R/Shiny ecosystem and the packages required for the interface, statistical processing and analysis, visualizations, modeling, and testing, including Shiny, bslib, testthat, and shinytest2. Packages specifically required by the statistical procedures and models will be managed using renv, facilitating consistent reproduction of the environment and its versions.

Project plan

Start-up phase

The project will begin by organizing the existing MVP code and defining the development and testing environment for the funded phase. The initial work will include establishing the project structure, confirming the required R/Shiny dependencies, configuring testthat and shinytest2, and defining the datasets and test scenarios to be used throughout the project.

Technical delivery

The funded work will be carried out over six months, organized into two three-month stages. Each stage will build on the results of the previous one: first, the current MVP will be stabilized and organized; subsequently, its existing functionalities will be strengthened.

Stage	Months	Main Activities	Results / Deliverables
1. Stabilization and Preparation	1–3	Optimize the workflows of the current MVP, strengthen code organization, configure testing with testthat and shinytest2, define datasets and test scenarios, and implement priority adjustments.	Stabilized MVP, consolidated modular structure, reproducible environment, and defined test scenarios, with initial adjustments implemented.
2. Strengthening and Improvements	4–6	Strengthen existing functionality, imputation options, diagnostics, model evaluation, results, and visualizations; expand testing and verification.	Strengthened functionalities, improved diagnostics and visualizations, expanded testing, and documented verification results.

Figure 2: Activity and Deliverable Timeline

Other aspects

The project’s code and development resources will be maintained in a public repository to support reuse and future contributions. An open-source license will be defined for the consolidated project. Public dissemination will be considered as part of the subsequent consolidation phase, including publication of project results and communication through R Consortium channels when appropriate.

Stage	Months	Main work	Hours	Requested Funds
1. Stabilization and Preparation	1–3	MVP workflow stabilization, code organization, testing setup, datasets and test scenarios, priority adjustments.	140	US\$4,200
2. Strengthening and Improvements	4–6	Strengthening existing functionalities, imputation, diagnostics, model evaluation, results, visualizations, testing and statistical verification.	193	US\$5,790
Total funded phase	1–6	Technical strengthening of the MVP	333	US\$9,990

Figure 3: Funded Work by Stage

Success

The main delivery risks relate to software stability, integration, and testing, and will be mitigated through progressive development, automated testing, QA, and systematic verification.

Definition of done

The funded phase will be considered complete when a technically strengthened, stable, tested, and reproducible MVP of Transparent Statistical Learning with R/Shiny is available. This version will include strengthened existing functionality, organized and reusable code, automated testing, statistical verification of relevant results, improved interface and visualizations, and a reproducible software environment.

Measuring success

Success will be measured based on the proper functioning of the main workflows, completion of the planned improvements, results of automated testing, and verification of selected statistical results. This will include functional and automated testing using reference datasets and scenarios, independent calculations, and verification of the reproducibility of the environment and analyses.

Future work

Subsequent phases may address comprehensive statistical and functional validation, expanded technical and user documentation, publication, translation, deployment, and further consolidation of the platform. The platform may also be progressively expanded through new statistical procedures, diagnostics, visualizations, documentation resources, and contributions from the R community. To facilitate reuse and extension, the project will provide a public repository, an open license, technical and user documentation, and reproducible resources.